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NORMALLY-OFF TRANSISTOR WITH REDUCED ON-STATE RESISTANCE AND MANUFACTURING METHOD

Abstract

A normally-off electronic device, comprising: a semiconductor body including a heterostructure that extends over a buffer layer; a recessed-gate electrode, extending in a direction orthogonal to the plane; a first working electrode and a second working electrode at respective sides of the gate electrode; and an active area housing, in the on state, a conductive path for a flow of electric current between the first and second working electrodes. A resistive region extends at least in part in the active area that is in the buffer layer and is designed to inhibit the flow of current between the first and second working electrodes when the device is in the off state. The gate electrode extends in the semiconductor body to a depth at least equal to the maximum depth reached by the resistive region.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a normally-off transistor with reduced ON-state resistance and to a method for manufacturing the transistor.

Description of the Related Art

[0002] Known to the art are high-electron-mobility transistors (HEMTs) with a heterostructure, made in particular of gallium nitride (GaN) and gallium and aluminum nitride (AlGaN). For instance, HEMT devices are appreciated for use as power switches thanks to their high breakdown threshold. Furthermore, the high current density in the conductive channel of the HEMT enables a low ON-state resistance ($R_{sub.ON}$) of the conductive channel to be obtained.

[0003] To favor use of HEMTs in high-power applications, HEMTs with normally-off channel have been introduced. HEMT devices with recessed-gate terminal have proven particularly advantageous for use as transistors with normally-off channel. A device of this type is, for example, known from Wantae Lim et al., “Normally-Off Operation of Recessed-Gate AlGaN/GaN HFETs for High Power Applications”, *Electrochem. Solid-State Lett.* 2011, volume 14, issue 5, H205-H207.

[0004] This HEMT has a gate trench that extends in depth in the heterostructure to the GaN layer. Extending in said trench is the gate metallization, which is separated from the AlGaN/GaN layers that form the heterostructure by a gate-dielectric layer. Formation of the gate trench is obtained by known steps of chemical etching and generates morphological defectiveness of various nature, such as for example even extensive surface corrugations or in general damage generated by the etching process (such as depressions or protuberances).

[0005] The U.S. Pat. No. 8,330,187 discloses a MOSFET with AlGaN/GaN heterojunction which has a recessed-gate terminal that extends in depth in a semiconductor body. The semiconductor body has, underneath the heterojunction, a GaN layer with a doping of a P-type, having the function of channel layer. Since the channel layer has a doping of a P-type, it makes it possible to obtain, in use, a transistor of a normally-off type with high turn-on threshold voltage. The gate terminal extends as far as the channel layer, and terminates within the channel layer itself. When, in use, the voltage applied to the gate terminal generates a charge-carrier inversion in the channel layer, a conductive channel is set up in the channel layer, which enables flow of a current between the source and drain terminals. However, the present applicant has found that the device according

to U.S. Pat. No. 8,330,187 has a high ON-state resistance due to the fact that the conductive channel is formed, for the most part, within the channel layer.

BRIEF SUMMARY

[0006] At least some embodiments provide a transistor of a normally-off type that enables a good trade-off between high threshold voltage and reduced ON-state resistance for overcoming the drawbacks of the known art.

[0007] According to at least some embodiments of the present disclosure a transistor of a normally-off type includes: [0008] a semiconductor body lying in a plane and including a buffer region and a heterostructure extending over the buffer region; [0009] a gate recessed electrode extending in the semiconductor body at least partially through the buffer region, along a direction orthogonal to the plane; [0010] a first working electrode and a second working electrode, which extend at respective sides of the gate electrode; and [0011] an active area which extends in the buffer region alongside and underneath the gate electrode and is configured to house, in a first operating condition in which a voltage between the gate electrode and the first working electrode is higher than a threshold voltage, a conductive path for a flow of an electric current between the first and second working electrodes.

[0012] The active area in the buffer region houses a resistive region configured to hinder, in a second operating condition in which the voltage between the gate electrode and the first working electrode is lower than the threshold voltage, the electric current flow between the first and second working electrodes. The resistive region extends at least in part in said active area, and said gate electrode extends in the semiconductor body to a depth, in said direction, equal to, or greater than, a maximum depth reached by the resistive region.

[0013] According to at least some embodiments of the present disclosure a method for manufacturing a normally-off transistor includes: [0014] forming a recessed gate electrode in a semiconductor body extending in a plane and including a buffer region and a heterostructure, the gate electrode extending at least partially through the buffer region, along a direction orthogonal to the plane; [0015] forming a first working electrode and a second working electrode at respective sides of the gate electrode, wherein the gate electrode, and the first and second working electrodes define an active area in the buffer region alongside and underneath the gate electrode, said active area being configured to house, in a first operating condition in which a voltage between the gate electrode and the first working electrode is higher than a threshold voltage, a conductive path for a flow of electric current between the first and second working electrodes; and [0016] forming a resistive region at least in part in the active area in said buffer region, said gate electrode extending in the semiconductor body as far as a depth, in the direction, equal to, or higher than, a maximum depth reached by the resistive region, the resistive region being configured to hinder, in a second operating condition in which the voltage between the gate electrode and the first working electrode is lower than the threshold voltage, the flow of electric current between the first and second working electrodes.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0017] For a better understanding of the present disclosure, preferred embodiments thereof are now described, purely by way of non-limiting example and with reference to the attached drawings, wherein:

[0018] FIG. 1 shows, in lateral section, a HEMT according to one embodiment of the present disclosure;

[0019] FIG. 2 shows, in lateral section, a HEMT according to a further embodiment of the present disclosure;

[0020] FIG. 3 shows, in lateral section, a HEMT according to a further embodiment of the present disclosure;

[0021] FIG. 4 shows, in lateral section, a HEMT according to a further embodiment of the present disclosure;

[0022] FIG. 5 shows, in lateral section, a HEMT according to a further embodiment of the present disclosure; and

[0023] FIGS. 6A-6E show steps for manufacturing the HEMT of FIG. 1.

DETAILED DESCRIPTION

[0024] FIG. 1 shows, in a triaxial system of orthogonal axes X, Y, Z, a HEMT device 1 of a normally-off type, based upon gallium nitride, including: a substrate 2, made, for example, of silicon, or silicon carbide (SiC), or sapphire (Al.sub.2O.sub.3); a buffer layer 11 extending over the substrate 2; and a heterojunction, or heterostructure, 7 extending over the buffer layer 11.

[0025] The buffer layer 11 comprises an electrical-conduction layer 4 and a resistive layer 6, where the electrical-conduction layer 4 is of gallium nitride (GaN) of an intrinsic type or with N-type doping and extends over the substrate 2, whereas the resistive layer 6 is of gallium nitride (GaN) with a doping of a P-type (for example, with a concentration of dopant species comprised between 10.sup.15 and 10.sup.20 ions/cm.sup.3) and extends over the electrical-conduction layer 4. The buffer layer 11 further comprises, optionally, one or more additional buffer layers (or interface layers) 3 of compounds formed by elements belonging to Groups III-V of the Periodic Table including gallium, which extend between the substrate and the electrical-conduction layer 4.

[0026] The buffer layer 11 has the function of configuring the device as a normally-off device.

[0027] The one or more interface layers 3 have the function of withstanding the drain voltage when the device is off and of decreasing the density of threading dislocations.

[0028] The heterostructure 7 includes, in particular, a barrier layer 9, made for example of gallium nitride (GaN) of an intrinsic type, extending over the resistive layer 6, and a channel layer 10, in this case of aluminum gallium nitride (AlGaN), extending over the barrier layer 9.

[0029] The HEMT device 1 further comprises an insulation layer 12, of dielectric material such as silicon nitride (Si.sub.3N.sub.4) or silicon oxide (SiO.sub.2), extending over a top side 7a of the heterostructure 7; and a gate region 14, extending between a source region 16 and a drain region 18.

[0030] In what follows, the substrate 2, the buffer layer 11 (and the buffer layer 3, when present), and the heterostructure 7 are referred to, as a whole, as the semiconductor body 15. The semiconductor body 15 houses an active region 15a, which forms the active part of the HEMT device 1.

[0031] The gate region 14 is separated laterally (i.e., along X) from the source region 16 and the drain region 18 by respective portions of the insulation layer 12. The gate region 14 is of a recessed type, and, according to one aspect of the present disclosure, extends in depth through the heterostructure 7, the resistive layer 6, and, in part, the electrical-conduction layer 4, terminating within the electrical-conduction layer 4. For example, considering an electrical-conduction layer 4 having a thickness, along Z, comprised between 20 nm and 10 μm, the gate region 14 extends in the electrical-conduction layer 4 for a depth greater than 0 μm and less than 10 μm, for example 0.5 μm.

[0032] According to a different aspect of the present disclosure, as illustrated in FIG. 2, the gate region 14 extends in depth right through the heterostructure 7 and the resistive layer 6 and terminates at the interface between the resistive layer 6 and the electrical-conduction layer 4. The gate region 14 thus reaches the electrical-conduction layer 4, but does not penetrate therein.

[0033] Irrespective of the embodiment, the gate region 14 is formed in a trench 19 etched through part of the semiconductor body 15. The trench 19 is partially filled by a dielectric layer 11, for example silicon oxide, which forms a gate-dielectric layer 14a. The gate-dielectric layer 14a extends over the bottom and inner side walls of the trench 19. A gate metallization 14b extends in

the trench **19** on the gate-dielectric layer **14a**. The gate-dielectric layer **14a** and the gate metallization **14b** form the gate region **14** of the HEMT device **1**.

[0034] According to further embodiments (not shown), the semiconductor body **15**, like the active region **15a** housed thereby, may comprise a single layer or a number of layers of GaN, or GaN alloys, appropriately doped or of an intrinsic type.

[0035] The source region **16** and drain region **18**, of conductive material, for example metal, extend over, and in contact with, the heterostructure **7**. According to a different embodiment, the source region **16** and drain region **18** may be of a recessed type, i.e., penetrate into a portion of the semiconductor body **15**.

[0036] The gate region **14** extends in an area corresponding to the active region **15a**.

[0037] In use, when the gate region **14** is biased with a voltage $V_{sub.G}$ higher than a threshold voltage $V_{sub.th}$, a conductive channel **22** is created (represented schematically by arrows) between the source region **16** and the drain region **18**, said channel extending in the direction Z through the resistive layer **6** and in the direction X through the electrical-conduction layer **4**, underneath the gate region **14**. In this way, the path of the current through the resistive layer **6**, of p-GaN, is minimized and the ON-state resistance, $R_{sub.ON}$, is optimized.

[0038] Operation of the HEMT device **1'** of FIG. **2** and the corresponding advantages are similar to those described with reference to the HEMT device **1** of FIG. **1**.

[0039] FIG. **3** shows a HEMT **30** according to a further embodiment of the present disclosure.

[0040] The HEMT **30** is similar to the HEMT **1** of FIG. **1** (elements that are in common are not described any further and are designated by the same reference numbers). However, in this case, the electrical-conduction layer **4** of GaN, represented in FIG. **1**, is replaced by an electrical-conduction layer **34** of a compound of gallium nitride comprising aluminum, such as AlGaN. In addition, extending between the electrical-conduction layer **4**, which is made, for example, of AlGaN, and the substrate **2**, is a gallium-nitride layer **35** for forming a further heterojunction, or heterostructure, **37** underneath the gate region **14**.

[0041] This solution is advantageous in so far as, in addition to the aforementioned advantages, the presence of the further heterostructure **37** underneath the gate region **14** enables formation of a layer of two-dimensional electron gas (2DEG) that reduces further the value of the ON-state resistance $R_{sub.ON}$ of the HEMT device **30**.

[0042] FIG. **4** shows a HEMT device **40** according to a further embodiment of the present disclosure.

[0043] The HEMT device **40** has, on the substrate **2** and on the buffer layer **3**, a heterostructure formed by a channel layer **44** and a barrier layer **46**. The channel layer **44** is, for example, of intrinsic gallium nitride (GaN), and the barrier layer **46** is, for example, of intrinsic aluminum gallium nitride (AlGaN). A gate region **48** of a recessed type extends between the source region **45** and the drain region **47**. The source region **45** and drain region **47** extend alongside the gate region **48**, on the barrier layer **46**. Optionally, also the source region **45** and drain region **47** may be of a recessed type. The channel layer **44** and the barrier layer **46** are of materials such that, when they are coupled together as illustrated in the figure, they form a heterojunction that enables formation of a 2DEG layer.

[0044] The gate region **48** extends, along Z, through the barrier layer **46** and the channel layer **44**, and terminates in the channel layer **44**.

[0045] A resistive region **50**, with P-type doping, extends alongside the gate region **48** and underneath the source region **45**. The resistive region **50** may extend both in the barrier layer **46** and in the channel layer **44**, or else exclusively in the channel layer **44**.

[0046] It may be noted that the resistive region **50** extends at least in part in lateral contact with the gate region **48** and completely underneath the source region **45** in such a way that, in use, the conductive channel is formed necessarily through it in order to enable a flow of electric current (represented by arrows **52**) between the source region **45** and the drain region **47**. The resistive

region **50** does not extend underneath the gate region **48**.

[0047] Optionally, a further resistive region (not represented in the figure) may be present on the opposite side of the gate region **48**, i.e., in lateral section, between the gate region **48** and the drain region **47** (in particular, specular to the resistive region **50**).

[0048] The resistive region **50** (and the further resistive region, when present) has a density of dopant species comprised between $10 \times 10^{15} \text{ cm}^{-3}$ and $10 \times 10^{20} \text{ cm}^{-3}$, for example $10 \times 10^{17} \text{ cm}^{-3}$.

[0049] According to a variant of the embodiment of FIG. **4**, the resistive region **50** extends (FIG. **5**) exclusively in the channel layer **44** at least in part in lateral contact with the gate region **48** and completely underneath the source region **45** in such a way that, also in this case, the conductive channel is formed through the resistive region **50**. A further resistive region (not shown) may be present on the opposite side of the gate region **48**, specular to the resistive region **50**.

[0050] Described in what follows, with reference to FIGS. **6A-6E**, are steps for manufacturing the HEMT device **1** of FIG. **1**.

[0051] FIG. **6A** shows, in cross-sectional view, a portion of a wafer **60** during a step of manufacture of the HEMT device **1**, according to one embodiment of the present disclosure. Elements of the wafer **60** that are in common to the ones already described with reference to FIG. **1** and appearing in FIG. **1** are designated by the same reference numbers.

[0052] In particular (FIG. **6A**), the wafer **60** is provided comprising: the substrate **2**, made, for example, of silicon (Si) or silicon carbide (SiC) or aluminum oxide (Al₂O₃), having a front side **2a** and a rear side **2b** opposite to one another in a direction Z; the electrical-conduction layer **4**, of intrinsic gallium nitride (GaN), having its own underside **4a** that extends on the front side **2a** of the substrate **2** (with the possible intermediate presence of the buffer layer **3**); the resistive layer **6**, of gallium nitride (GaN) with P-type doping; and the heterostructure **7**, extending over the resistive layer **6**.

[0053] By way of example, the resistive layer **6** has a thickness comprised between 5 nm and 1 μm , and the GaN layer **10** has a thickness comprised between a few nanometers (e.g., 2 nm) and 1 μm .

[0054] According to the present disclosure, extending on the front side **7a** of the heterostructure **7** is the passivation layer, or insulation layer, **12**, of dielectric or insulating material such as silicon nitride (SiN), silicon oxide (SiO₂), or some other material still. The insulation layer **12** has a thickness comprised between 5 nm and 300 nm, for example 100 nm, and is formed by chemical-vapor deposition (CVD) or atomic-layer deposition (ALD).

[0055] The wafer **60** according to FIG. **6A** may be purchased prefabricated or else formed by processing steps in themselves known.

[0056] Next (FIG. **6B**), the insulation layer **12** is selectively removed, for example with lithographic and etching steps, for removing selective portions thereof in the region of the wafer **60** where, in subsequent steps, a gate region of the HEMT device is to be formed (i.e., in an area corresponding to a part of the active area **15a**).

[0057] The etching step may stop at the electrical-conduction layer **4** (in a way not represented in the figure), or else proceed partially into the electrical-conduction layer **4** (the latter solution is represented in FIG. **6B**). In either case, a surface portion **4'** of the electrical-conduction layer **4** is exposed. The portion of the electrical-conduction layer **4** removed generates a cavity, in the electrical-conduction layer **4**, having a depth d_i along Z comprised between 0 and 1 μm . However, other embodiments are possible, and the portion of the electrical-conduction layer **4** removed may have a depth, along Z, greater than 1 μm (in any case less than the total thickness of the electrical-conduction layer **4**).

[0058] Then (FIG. **6C**), a step is carried out of deposition, or growth, of the gate-dielectric layer **14a**, for example of a material chosen from among aluminum nitride (AlN), silicon nitride (SiN), aluminum oxide (Al₂O₃), and silicon oxide (SiO₂). The gate-dielectric layer **14a** has a thickness chosen between 5 nm and 50 nm, for example, 20 nm.

[0059] Next (FIG. 6D), a step is carried out of deposition of conductive material on the wafer **60** to form a conductive layer **58** on the gate-dielectric layer **14a**, in particular filling the trench **19** completely. For example, the conductive layer **58** is of metal material, such as tantalum (Ta), tantalum nitride (TaN), titanium nitride (TiN), palladium (Pa), tungsten (W), tungsten silicide (WSi.sub.2), titanium aluminum (Ti/Al), and nickel gold (Ni/Au).

[0060] The conductive layer **58** is then selectively removed with lithographic and etching steps in themselves known for eliminating the conductive layer **58** from the wafer **60** except for the portion thereof that extends in the trench **19** to form the gate metallization **14b**. The gate metallization **14b** and the gate dielectric **14a** form, as a whole, the recessed-gate region **14** of the HEMT device **1** of FIG. **1**.

[0061] Then (FIG. 6E), one or more further steps are carried out of masked etching of the dielectric layer **14a** and of the insulation layer **12**, to remove selective portions thereof that extend in regions of the wafer **60** where the source and drain regions **16**, **18** of the HEMT device **1** are to be formed.

[0062] In particular, openings **54a** and **54b** are formed on opposite sides, along X, of the gate region **14**, and at a distance from the gate region **14**.

[0063] Next, a step of formation of ohmic contacts is carried out to provide the source and drain regions **16**, **18**, depositing conductive material, in particular metal such as titanium (Ti) or aluminum (Al), or their alloys or compounds, by sputtering or evaporation, on the wafer **60** and in particular inside the openings **54a**, **54b**. There is then carried out a subsequent step of etching of the metal layer thus deposited to remove said metal layer from the wafer **60** except for the metal portions that extend within the openings **54a** and **54b**, to form in said openings **54a** and **54b** the source region **16** and the drain region **18**, respectively.

[0064] Then, a step of rapid thermal annealing (RTA), for example at a temperature between approximately 500° C. and 900° C. for a time ranging from 20 s to 5 min, enables formation of ohmic contacts of the source region **16** and drain region **18** with the underlying heterostructure **7**.

[0065] The HEMT device **1** represented in FIG. **1** is thus formed.

[0066] With reference to the embodiment of FIG. **3**, the manufacturing steps are similar to the ones described with reference to FIGS. **6A-6E**, with the difference that, as an alternative to the electrical-conduction layer **4**, of GaN, the layers **35** and **34**, made, respectively, of GaN and AlGaIn, are formed, stacked on top to one another.

[0067] With reference to the embodiment of FIG. **4**, in this case, after providing a wafer comprising the substrate **2**, the channel layer **44**, and the barrier layer **46**, prior to formation of the gate region **48**, source region **45**, and drain region **47**, a step of implantation of dopant species, for example Mg, Zn, F, is carried out using as parameters an implantation energy of 30 keV and an implantation dose of 10.sup.15 cm.sup.-2. The insulation layer **12** may be present during implantation in order to limit surface damage of the wafer. A step of thermal annealing enables activation of the implanted dopant species to form the resistive region **50** of FIG. **4**.

[0068] By modulating the implantation energy, it is possible to modulate the implantation depth. For example, by increasing the implantation energy it is possible to form the resistive region **50** exclusively in the channel layer **44**, at the desired depth. Use of an implantation step enables, in particular, definition of the resistive region only in the low-field region of the device.

[0069] The implantation steps are carried out using an appropriate mask in order to define the extension, in the plane XY, of the resistive implanted region.

[0070] The advantages of the disclosure according to the present disclosure emerge clearly from what has been set forth previously.

[0071] In particular, a considerable improvement of the trade-off between turn-on threshold voltage (V.sub.th) and ON-state resistance (R.sub.ON) is obtained.

[0072] Finally, it is clear that modifications and variations may be made to what has been described and illustrated herein, without thereby departing from the scope of the present disclosure.

[0073] For example, at the interface between the substrate **2** and the electrical-conduction layer **4**

there may be present one or more further transition layers (not shown) of gallium nitride and compounds thereof, such as, for example, AlGaN, or AlN, having the function of interface for reducing the lattice misalignment between the substrate **2** and the electrical-conduction layer **4**. [0074] The metallization of the contacts (source, drain, gate) on the front of the wafer may be carried out using any variant known in the literature, such as, for example, formation of contacts of AlSiCu/Ti, Al/Ti, or W-plugs, or others still.

[0075] The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. A method, comprising: forming an insulation layer on a semiconductor body, the semiconductor body including a buffer region and a heterostructure on the buffer region, the insulation layer having a first surface; forming a trench extending through the insulation layer and at least partially through the buffer region along a first direction; forming a gate dielectric layer in the trench and on the first surface of the dielectric layer; forming a gate electrode in the trench, the gate electrode extending into the semiconductor body and at least partially through the buffer region; forming a source electrode and a drain electrode, the gate electrode disposed between the source and drain electrodes along a second direction that is transverse to the first direction, the gate electrode, and the source and drain electrodes defining an active area in the buffer region, the active area having a depth extending along the first direction that is greater than a depth of the gate electrode, and having a width extending along the second direction between the source electrode and the drain electrode, wherein each of the gate electrode, the source electrode, and the drain electrode includes respective portions which extend over the first surface of the insulation layer along the first direction.
2. The method of claim 1, wherein forming the gate electrode and forming the source electrode and the drain electrode includes forming the gate electrode, the source electrode, and the drain electrode on the gate dielectric layer, the gate dielectric layer extending along the first direction directly between the first surface of the insulation layer and the respective portions of the gate electrode, the source electrode, and the drain electrode.
3. The method of claim 1, further comprising: forming a resistive layer, the active area in the buffer region including the resistive layer, wherein the active area is configured to, in a first operating condition in which a first voltage between the gate electrode and one of the source electrode or the drain electrode is higher than a threshold voltage, provide a conductive path for a flow of an electric current between the source and drain electrodes, and wherein the resistive layer is configured to, in a second operating condition in which a second voltage between the gate electrode and the one of the source electrode or the drain electrode is lower than the threshold voltage, hinder an electric current flow between the source and drain electrodes.
4. The method of claim 3, wherein forming the resistive layer includes implanting dopant species of a P-type between the gate electrode and at least one of the source electrode or the drain electrode.
5. The method of claim 3, wherein forming the resistive layer includes depositing a layer of a compound formed by elements of Groups III-V with a doping of a P-type.
6. The method of claim 5, further comprising forming an electrical-conduction layer in the active area in the buffer region, the electrical-conduction layer formed by elements of Groups III-V of an intrinsic type or with a doping of N-type, the resistive layer disposed on the electrical-conduction

layer.

7. The method of claim 1, wherein forming the source electrode and the drain electrode includes: forming openings extending through the insulation layer; depositing a conductive material in the openings; and forming ohmic contacts of the source electrode and drain electrode with the heterostructure by performing a rapid thermal annealing.

8. A method, comprising: forming a trench in a stack of layers, the trench extending along a first direction through an insulation layer, a heterostructure, and a resistive layer; forming a gate dielectric layer on the insulation layer and in the trench, the gate dielectric layer covering a first surface of the trench and a plurality of sidewalls of the trench; forming a conductive layer in the trench and on the insulation layer; and forming a source electrode and a drain electrode in the stack of layers, the source and drain electrodes each extending along the first direction through the gate dielectric layer and the insulation layer.

9. The method of claim 8, wherein the trench extends partially through an electrical-conduction layer along the first direction.

10. The method of claim 8, wherein the stack of layers includes an electrical-conduction layer on a substrate, the resistive layer on the electrical-conduction layer, the heterostructure on the resistive layer, and the insulation layer on the heterostructure.

11. The method of claim 10, wherein the heterostructure includes a barrier layer on a channel layer.

12. The method of claim 9, wherein the trench extends a first distance into the electrical-conduction layer along the first direction, the first distance being in the range of 0 and 1 μm .

13. The method of claim 8, wherein the conductive layer entirely covers the gate dielectric layer on the insulation layer, on the first surface of the trench, and on the plurality of sidewalls of the trench.

14. The method of claim 8, comprising forming a gate metallization by removing a first portion of the conductive layer.

15. The method of claim 8, wherein the trench is between the source and drain electrodes along a second direction transverse to the first direction.

16. A method, comprising: forming, in a stack of layers on a substrate, a trench extending along a first direction through an insulation layer, a heterostructure, and at least partially through a buffer layer; forming a gate dielectric layer on the insulation layer and in the trench, the gate dielectric layer covering a first surface of the trench and a first sidewall of the trench; forming a gate electrode having a first portion in the trench and a second portion outside the trench, the first portion having a first width along a second direction transverse to the first direction and the second portion having a second width along the second direction greater than the first width; and forming a source electrode and a drain electrode in the stack of layers, the gate electrode being between the source and drain electrode along the second direction, the source and drain electrodes each extending along the first direction through the insulation layer.

17. The method of claim 16, wherein the buffer layer includes a resistive layer on an electrical-conduction layer, the trench extending entirely through the resistive layer.

18. The method of claim 17, wherein the trench extends at least partially through the electrical-conduction layer.

19. The method of claim 16, wherein the second portion of the gate electrode extends over the insulation layer along the second direction.

20. The method of claim 19, wherein the source and drain electrodes each include a first portion in the insulation layer and a second portion outside of the insulation layer, the second portion of both the source and drain electrodes extending over the insulation layer along the second direction.
